Step-by-Step Build Guide: Water Pipe Leak Detection Model

Follow the steps in order. Each step lists the materials it needs, what to do and how to check that it worked. Total build time is about 2 days, and the cost is about ₹3,350.

Before you start

Tools needed

- Small screwdriver set, wire stripper, scissors or pipe cutter, hacksaw (for PVC)
- Drill with 4 mm bit (for mounting holes)
- Hot glue gun or double-sided tape
- Digital multimeter (very useful for checking 5V and 12V)
- Soldering iron and solder (optional, for the final neat version)
- Laptop with the Arduino IDE installed

Complete materials list

No.	Material	Qty	Cost (₹)
1	Arduino Uno or Nano, with USB cable	1	450
2	Water pressure sensor, 0 to 1.2 MPa, 5V, G1/4 thread (S1, S2)	2	900
3	12V DC diaphragm pump	1	350
4	Half-inch PVC pipe, about 1.5 m	1	150
5	Fittings: 3 tees, 2 elbows, reducer bushings (half inch to quarter inch BSP) for the sensors	6	200
6	Small ball valve or tap (leak tap)	1	100
7	16x2 LCD with I2C module	1	180
8	Buzzer, 1 red LED, 1 green LED, two 220 ohm resistors	1 set	50
9	5V single-channel relay module	1	60

No.	Material	Qty	Cost (₹)
10	12V solenoid valve, normally closed	1	350
11	12V 3A adapter with DC jack	1	250
12	Breadboard, jumper wires (male-male, male-female), 1N4007 diode, ON/OFF switch	1 set	150
13	Plywood or acrylic base board, about 60 x 40 cm	1	100
14	Water tank (bucket or plastic box) and a tray for the leak	1	50
15	Silicone tubing (about 8 mm), PTFE tape, PVC solvent cement, cable ties, clamps	1 set	100

Before buying all the fittings, check your pump's outlet size and the sensor thread size. Pumps and sensors differ, so match the adapters to what you actually have.

Step 1: Prepare the base board and plan the layout

Materials: base board, pencil, ruler, tank, pump, pipe pieces (not glued yet).

1. Place the board flat on a table.
2. Put the tank at the left end and the pump next to it.
3. Lay the pipe in a straight run to the right, with space for two sensors and the leak tap along it.
4. Keep the bottom third of the board free for the electronics (Arduino, breadboard, relay, LCD).
5. Mark the positions with a pencil. Use the 3D layout page from this project as a reference.

Check: nothing overlaps and every part has a place.

Step 2: Build the water loop (pipe section)

Materials: half-inch PVC pipe, 3 tees, 2 elbows, hacksaw, PVC solvent cement, PTFE tape, silicone tubing, clamps.

1. Cut the PVC pipe into pieces: one 15 cm piece from the pump to the first tee, one 40 cm piece between the first and second tee, one 15 cm piece to the third tee, and a short piece for the outlet.
2. Dry-fit everything first without glue.
3. Order along the pipe: pump outlet, tee for S1, 40 cm section with a tee for the leak tap in the middle, tee for S2, solenoid valve, return tubing to the tank.
4. When the layout is right, glue the PVC joints with solvent cement and hold each joint for 30 seconds.
5. Connect the pump outlet to the pipe with a barbed or threaded adapter and silicone tubing, fixed with a clamp.
6. Let the cement cure for at least 2 hours.

Check: the pipe is straight, joints are firm and the pipe is fixed to the board with clamps or cable ties.

Step 3: Install the pressure sensors S1 and S2

Materials: 2 pressure sensors, 2 reducer bushings, PTFE tape.

1. Wrap 5 to 6 turns of PTFE tape clockwise around the sensor thread.
2. Screw each sensor into the side port of its tee (through the reducer bushing). Hand-tight plus a quarter turn is enough. Do not over-tighten.
3. S1 goes on the first tee (near the pump), and S2 goes on the tee at the far end of the monitored section.
4. Point the sensors upward so air bubbles do not collect against the sensor face.
5. Fix the sensor bodies to the board with a cable tie so the thread is not stressed.

Check: both sensors are firm and the thread is wrapped in tape.

Step 4: Install the leak tap

Materials: small ball valve or tap, PTFE tape, tray or bucket.

1. Screw the tap into the middle tee of the 40 cm section, pointing downward, with PTFE tape on the thread.
2. Place a tray or bucket under it to catch the water.
3. The tap stays closed during normal running. Opening it simulates a leak.

Check: the tap opens and closes smoothly and the tray sits under it.

Step 5: Install the solenoid valve

Materials: 12V normally closed solenoid valve, fittings or tubing to fit it, PTFE tape, clamps.

1. Fit the valve after S2, so it controls the flow leaving the monitored section. Check the arrow on the valve body and install it in the arrow direction.
2. Connect its outlet with tubing back to the tank.
3. Do not connect any wires yet.

Check: water has a clear path from pump to tank through the valve.

Step 6: Water test with no electronics

Materials: water, tank, 12V adapter, ON/OFF switch, wire, towel.

1. Fill the tank. Connect the pump to the 12V adapter through the ON/OFF switch. Leave the Arduino and sensors disconnected.
2. To test the loop without the valve (it is closed with no power), temporarily connect 12V directly to the valve so it opens.
3. Switch the pump on and watch every joint for 5 minutes.
4. Fix any drip by drying the joint and re-taping or re-gluing it.
5. Open the leak tap and confirm water flows out into the tray, then close it.

Check: no leaks anywhere except through the leak tap.

Step 7: Set up the Arduino and LCD

Materials: Arduino, USB cable, breadboard, LCD I2C module, 4 jumper wires, laptop.

1. Install the Arduino IDE. Go to Sketch, Include Library, Manage Libraries, and install LiquidCrystal I2C by Frank de Brabander.
2. Connect the LCD: GND to GND, VCC to 5V, SDA to A4, SCL to A5.
3. Upload this short test sketch:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup() { lcd.init(); lcd.backlight(); lcd.print("Hello"); }
void loop() {}
```

4. If the screen is blank, turn the small blue contrast screw on the back of the LCD. If it is still blank, change 0x27 to 0x3F.

Check: the LCD shows the word Hello.

Step 8: Connect the sensors and read raw values

Materials: 2 sensors (already fitted), jumper wires, breadboard.

1. With power off, wire each sensor: red to 5V, black to GND, yellow (signal) to A0 for S1 and A1 for S2.
2. Upload this sketch and open Tools, Serial Monitor at 9600 baud:

```
void setup() { Serial.begin(9600); }
void loop() {
  Serial.print(analogRead(A0)); Serial.print("  ");
  Serial.println(analogRead(A1));
  delay(500);
}
```

3. With the pump off, both numbers should be around 100 to 105 (this is the 0.5 V zero level).
4. Switch the pump on. Both numbers should rise and stay close together.

Check: both sensors respond to the pump and read similar values. If a number is stuck at 0 or 1023, recheck that sensor's wiring.

Step 9: Add the LEDs and buzzer

Materials: red LED, green LED, two 220 ohm resistors, buzzer, jumper wires.

1. Red LED: Arduino D9, through a 220 ohm resistor, to the LED long leg. The short leg goes to GND.
2. Green LED: D10, through a 220 ohm resistor, to its long leg. The short leg goes to GND.
3. Buzzer: positive to D8, negative to GND.

4. Quickly test each with `digitalWrite(pin, HIGH)` in a test sketch.

Check: both LEDs light and the buzzer sounds.

Step 10: Wire the relay and the solenoid valve

Materials: 5V relay module, solenoid valve, 12V adapter, 1N4007 diode, jumper wires.

1. Relay module low side: IN to D7, VCC to 5V, GND to GND.
2. Relay high side: connect the adapter's +12V to the relay COM terminal. Connect the relay **NC** terminal to the valve's positive wire. Connect the valve's other wire to the adapter's GND.
3. Fit the 1N4007 diode across the valve wires, with the stripe on the +12V side. It absorbs voltage spikes when the valve switches off.
4. Do the wiring with the adapter unplugged.
5. How it works: with the usual active-low relay module, the relay stays off during normal running, so the valve is powered and open. On a leak the Arduino turns the relay on, which cuts power and the valve closes.
6. Test: plug in the adapter. The valve should be open (you hear a click on power-up). Set D7 LOW and the relay should click and the valve should close.

Check: the valve opens and closes with the relay. If it works the opposite way, your relay module is active-high, so swap HIGH and LOW in the code.

Step 11: Upload the full program

Materials: Arduino, laptop, USB cable.

Upload this complete sketch:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);   // try 0x3F if blank

const int S1 = A0, S2 = A1;
const int BUZZER = 8, RED = 9, GREEN = 10, RELAY = 7;

float zero1 = 0, zero2 = 0;
const float LEAK_DIFF_KPA = 15.0;      // tune by testing
```

```
const unsigned long CONFIRM_MS = 3000; // leak must last 3 s
unsigned long leakStart = 0;
bool leakLatched = false;

float readKPA(int pin) {
  long sum = 0;
  for (int i = 0; i < 20; i++) { sum += analogRead(pin); delay(2); }
  float v = (sum / 20.0) * 5.0 / 1023.0;
  float mpa = (v - 0.5) / 4.0 * 1.2;   // 0.5 to 4.5 V = 0 to 1.2 MPa
  return mpa * 1000.0;                 // kPa
}

void setup() {
  Serial.begin(9600);
  pinMode(BUZZER, OUTPUT); pinMode(RED, OUTPUT);
  pinMode(GREEN, OUTPUT);  pinMode(RELAY, OUTPUT);
  digitalWrite(RELAY, HIGH);           // relay off: valve powered and open
  lcd.init(); lcd.backlight();
  lcd.print("Calibrating...");
  delay(2000);                         // pump must be OFF now
  zero1 = readKPA(S1);
  zero2 = readKPA(S2);
  lcd.clear();
}

void loop() {
  float p1 = readKPA(S1) - zero1;
  float p2 = readKPA(S2) - zero2;
  float diff = p1 - p2;

  if (p1 > 20 && diff > LEAK_DIFF_KPA) {
    if (leakStart == 0) leakStart = millis();
    if (millis() - leakStart > CONFIRM_MS) leakLatched = true;
  } else {
    leakStart = 0;
  }

  lcd.setCursor(0, 0);
  lcd.print("P1:"); lcd.print(p1, 0); lcd.print(" P2:"); lcd.print(p2, 0);
lcd.print("  ");
  lcd.setCursor(0, 1);
  lcd.print(leakLatched ? "LEAK DETECTED!  " : "Pipe OK         ");
```

```
  digitalWrite(RED, leakLatched);
  digitalWrite(GREEN, !leakLatched);
  digitalWrite(BUZZER, leakLatched);
  digitalWrite(RELAY, leakLatched ? LOW : HIGH);  // LOW = relay on = valve
closed

  Serial.print(p1); Serial.print(","); Serial.println(p2);  // for Serial
Plotter
  delay(200);
}
```

Check: with the pump off the LCD shows Calibrating, then P1 and P2 near 0 and Pipe OK.

Step 12: Calibrate and run the first test

Materials: assembled model, water, laptop.

1. Always power the Arduino with the pump OFF so it records the zero reading.
2. After the LCD shows Pipe OK, switch the pump on.
3. P1 and P2 should rise (roughly 100 to 300 kPa for a small pump) and stay within a few kPa of each other.
4. Watch for one minute. If the alarm triggers on its own, increase `LEAK_DIFF_KPA` (for example to 20) and re-upload.

Check: steady readings and Pipe OK for one minute, with the green LED on.

Step 13: Leak test

Materials: assembled model, water, tray.

1. Open the leak tap.
2. P2 should fall below P1. After the difference stays above the threshold for 3 seconds, the red LED turns on, the buzzer sounds, the LCD shows LEAK DETECTED and the valve closes.
3. Close the leak tap, switch the pump off, then press the Arduino reset button. Switch the pump on only after the LCD shows Pipe OK again.
4. Repeat with a small, medium and fully open tap and write down P1, P2 and the difference each time. This is your data for the report.

Important: when the valve closes, the pump pushes against a closed line. A small diaphragm pump can usually take this for a short time, but switch it off soon after the alarm. A pump with a built-in pressure switch is safer.

Check: the alarm and valve action work every time, and a closed tap never causes a false alarm.

Step 14: Add the live graph on the laptop

Materials: laptop, USB cable (the code from step 11 already sends the data).

1. Close the Serial Monitor.
2. Open Tools, Serial Plotter and set the speed to 9600 baud.
3. You see two lines, P1 and P2. During a leak P2 drops away from P1.
4. Take a screenshot of normal running and of a leak for your presentation.

Check: the graph updates live and shows the gap during a leak.

Step 15: Finish and tidy the model

Materials: cable ties, labels or marker, hot glue, optional small box for the electronics.

1. Move the circuit from the breadboard to a neat layout (or solder it on a perfboard if you have time).
2. Fix the Arduino, relay and LCD to the board with screws, double-sided tape or hot glue.
3. Tie all wires neatly with cable ties and keep the 12V wires away from the sensor wires.
4. Label S1, S2, the leak tap, the valve, the buzzer and the LEDs.
5. Add a printed title and a block diagram next to the model.
6. Keep the electronics away from the leak tap and the tank. Cover any exposed connections.

Check: the model looks clean and nothing can touch water by accident.

Step 16: Demo script (2 minutes)

1. Say the problem: pipes leak and the loss goes unnoticed.
2. Show the model and name the parts: pump, S1, S2, leak tap, valve, Arduino, laptop.
3. Power up with the pump off, so the system calibrates. Then switch the pump on and show Pipe OK and a flat graph.

4. Open the leak tap. Point to the falling P2 on the graph, then to the alarm, the LCD message and the valve closing.
5. Close the tap, reset, and show the system back to normal.
6. Say that the system detects the leak and stops the water loss automatically, and the physical repair is done by a person. Mention the future scope: more sensor pairs, flow sensor and phone alerts.

Troubleshooting

Problem	Likely cause	Fix
LCD is blank	Wrong I2C address or contrast	Try 0x3F and turn the contrast screw
Sensor reads 0 or 1023	Wrong or loose wiring	Recheck red, black and yellow wires
P1 and P2 differ even with no leak	Zero offset or air in the pipe	Calibrate with the pump off and let air bubbles out
False alarms	Threshold too low or pump pulsing	Raise `LEAK_DIFF_KPA` or increase `CONFIRM_MS`
No alarm when the tap is open	Threshold too high or leak too small	Lower the threshold or open the tap more
Valve works backwards	Relay module is active-high	Swap HIGH and LOW for the relay
Relay clicks but valve does not move	Valve on NO instead of NC, or no 12V	Use the NC terminal and check 12V with the multimeter
Water drips at a joint	Poor sealing	Dry the joint and re-wrap with PTFE tape

Safety notes

- Keep all mains-powered parts and the adapter away from water.
- Unplug the 12V adapter before changing the wiring.
- Never exceed the sensor pressure rating of 1.2 MPa.

- Use a tray under the model to catch spills.